



Constructing a Guangdong-Hong Kong-Macau Greater Bay Area social space: cross-sectoral experiences of key node university faculties

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Abstract

This paper explores how university faculties with experience of brokering academic, industrial, and governmental networks, as key nodes, perceive, conceive, and live the Guangdong-Hong Kong-Macau Greater Bay Area (GBA) higher education space. Drawing on Henri Lefebvre's (1991) critical work on the production of space and Ronald S. Burt's (2000) social network theory, this study draws primarily on semi-structured interviews with faculties from Guangdong, Hong Kong and Macau Special Administrative Regions, complemented with policies, investment statistics, and funding schemes. It suggests that the GBA space attains "real" existence by virtue of networks, pathways, and clusters of relationships built within and across its triad of space, particularly the "structural holes" addressed by key node faculties. These early strategic movers create a new equilibrium of demand for the GBA through cross-fertilizing networks of academia, industry, and government across city-regions and administrative borders. This study contributes theoretically and methodologically by combining spatial and social network theories to unpack the under-examined "lived" space, which is often seen as subordinate to "perceived" and "conceived" spaces. It provides significant policy and practical implications for optimizing regulations to better coordinate resources and institutions supportive of multi-sectoral networks and practices across the regional development of the GBA and beyond.

Keywords Higher education · Society interaction · Cross-sectoral experiences · Network structure · Key nodes

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Introduction

Higher education has been positioned as a key driver for regional development by fostering highly qualified manpower, developing regional capacities for technology transfer and innovation, and building regional identity and reputation (Callender et al., 2020; Chankseliani et al., 2021; Etzkowitz, 2004; Faggian et al., 2019; Tang, 2022). Academic entrepreneurship through spin-offs becomes a source of new firms, revenues, and jobs (Power & Malmberg, 2008). This role has expanded amid growing emphasis on systematic and scientific approach to policymaking involving academia and industry, alongside the rise of entrepreneurial reforms and academic capitalism aimed at aligning higher education with state, market, and societal demands (Slaughter, & Rhoades, 2004).

In China, the promotion of subnational higher education regionalization initiative and entrepreneurial reforms has also been rhetorically embraced as strategies for modernization and excellence and implemented with initiatives for technology transfer and academic entrepreneurship (Zhu & Yang, 2024b). Both central and provincial governments have identified the strategic role of higher education in transforming the manufacturing model into one primarily driven by scientific and technological innovation in the GBA, which was established in the *Outline Development Plan for the GBA* (State Council, 2019), with the aim of strengthening regional synergy and becoming a global innovation cluster despite the “one country, two systems, and three tariff zones” of Guangdong Province, and the Hong Kong and Macau Special Administrative Regions. Numerous policies and institutions are in place to incentivize and support collaborations across sectors and city-regions.

Amid the building of multi-scalar networks across academia, industry, and government, university faculties serve as the agents and nodes that mobilize ideas, knowledge, resources, rules, and power across institutional and city-region boundaries. As “network brokers” that address structural holes (i.e., the separation of groups across space (Burt, 2000)), they create space for collaborative agenda-setting, knowledge production, and institutional innovation, drawing more nodes to existing networks to enhance their breadth and density. The role of academics as key advisors for industry and government is also enhanced by the increasingly blurred interface of economic development, public policy, and academic knowledge (Christensen & Mandelkern, 2022; Han & Ye, 2017).

Extant research on the GBA initiative has largely framed its success in terms of infrastructure integration and policy coordination (Lo & Li, 2023). Despite policy-driven efforts, the construction of a regional higher education space through cross-border and cross-sectoral network-building remains fragmentary, superficial, and fraught with implementation ambiguities (Marginson & Considine, 2000; Zhu & Yang, 2024b). Yet how integration is defined, enacted, and by whom, beyond the dominant policy narrative, remains under-explored.

Has the GBA higher education produced a space of its own? How do academics, the very agents that the regional strategy strive to mobilize, perceive, conceive, and live within this space? These questions remain critical, as policies that fail to generate new spaces fall short of transforming lived experiences, merely reshaping the “ideological superstructure” or “political apparatuses” (Lefebvre, 1991, p. 54). Hence, situated within the “spatial problematic” (Soja, 1985), this study aims to explore the experiences of university faculties who take on multiple roles, connecting the networks of academia, industry and government based on interviews and supplementary evidence from infrastructure and funding statistics.

Literature review

Amid globalization and international competition in the knowledge economy, there are growing societal and institutional expectations for enhancing the economic role of knowledge and innovation through function complementarity of academia, industry, and government and the synergistic value of hybrid logics and capacities to produce a self-perpetuating ecosystem (Etzkowitz & Dzisah, 2008; Etzkowitz & Ranga, 2010).

Academic entrepreneurship is gaining visibility as a source of commercialization of knowledge (Shore & McLauchlan, 2012), a creator of new firms, revenues, and jobs (Power & Malmberg, 2008) and as a new material force in the region and a vehicle for regional technology transfer (Benneworth & Charles, 2005; Chiesa & Piccaluga, 2000; Shane, 2004). It has become a major strand of academic cross-sectoral practices, encompassing both formal activities such as spin-outs, licensing, and joint ventures and informal knowledge transfer, including consultancy, contract and collaborative research, shared facilities, secondments, training, professional development, and student placements (Ankrah & Omar, 2015). These diverse practices generate significant economic and social value for both academics and external partners.

Motivations for engagements are shaped by a mix of institutional, intrinsic, reputational, financial, and research-related goals such as advancing scholarly agendas, gaining recognition, and demonstrating societal relevance while financial incentives tend to be secondary or uneven in influence (Belkhodja & Landry, 2007; D'este & Perkmann, 2011; Orazbayeva et al., 2020). These individual drivers align with the institutional imperatives of the entrepreneurial university, which seeks to enhance external impact, diversify funding streams, and position academic work within broader innovation and economic development agendas (Abreu & Grinevich, 2013).

Alongside notable interests in promoting academia-industry collaboration and academic entrepreneurship for fostering innovative regions, the state-of-the-art and policy agendas on knowledge exchange and impact have also highlighted broader engagement with the government, market, and society (Koekkoek et al., 2021; Perkmann et al., 2021). There has been growing momentum to integrate research evidence into public policymaking by involving scholars in expert panels for government-funded projects and promoting policy engagement through secondments and career pathways within government agencies to enhance the scientific rigour of policymaking (UK Government, 2025).

Many academics are active in civic engagement as public consultants and advisory experts, drafting relevant documents and resources, designing transformative socio-political agendas, service and training programs, or local initiatives in building public infrastructures (Luke et al., 2013; Soska & Butterfield, 2013; Strier, 2014; Kindred & Petrescu, 2015; Sasson, 2019; Groulx et al., 2021). The interstitial space between academia and government is often occupied by a small group of elite individuals who traverse sectors, facilitate knowledge exchange, and actively circulate, interpret, and translate scientific knowledge to make it socially and politically meaningful across policy domains and public debates (Thune et al., 2023).

This scholarship–government linkage is even more pronounced and nuanced in the Chinese context and arguably also in the GBA. As Zhu and Yang (2024b) noted, academia–industry engagement is underpinned by patriotic ideals such as devotion to family and nation while serving the nation through industrial development. Government influences academic activity through policy instruments and market mechanisms, with state and market forces reinforcing one another to drive academic productivity. Counter to the

“bottom heaviness” of collegiality of academics that are in part conservative and resistant to change in entrepreneurial universities (Clark, 2001, p.22), Zhu and Yang (2024b) indicated that academics remain largely subordinated to the “top heaviness” of central party-state, though with intellectual autonomy to align individual interest with government objectives.

Nevertheless, tensions, and at times contradictory priorities, persist within the unbalanced academic–government–market dialogue. Across Chinese and broader contexts, it remains an open question how to address concerns that entrepreneurial reforms and Triple Helix models have intensified government and market influence at the expense of academic agency, which is further complicated by the GBA’s heterogeneous institutional landscape and the interplay of Western and Chinese norms shaping academic agency in cross-sectoral engagement (Zhu & Yang, 2024b).

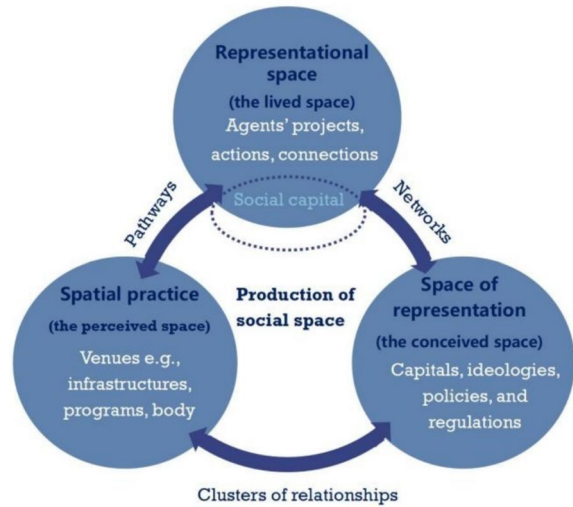
While the intermediary institutional level to support faculties’ entrepreneurial practices is yet to develop in China as well as in the GBA (Zhu & Yang, 2024a), the bottom-up agency of influential individual academics plays a critical but under-examined role in bridging cross-sectoral networks, synthesizing institutional logics and resources, and integrating the spaces of knowledge, production, and politics (Etzkowitz & Ranga, 2012). With crucial visions, resources, and “skills in playing power games,” some even act as “institutional entrepreneur” exerting greater leeway to interpret rules, negotiate with existing structures, and balance compliance and innovation (Abdelnour et al., 2017, p. 1776; Ma & Cai, 2021, p. 912).

Existing studies of higher education in the GBA focus predominantly on outlining the policy and infrastructure landscape and presenting opportunities to higher education, which also comes with normative expectation for higher education to work closer with the industry and contribute more significantly to the integration, innovation and internationalization of the region (Mok, 2022; Xie et al., 2021). Yet scant attention has been paid to the granularity of university faculties, the actual agents living the higher education space and nodes that connect institutional and regional networks across the GBA. Although a few studies employ quantitative mappings of technology transfer and knowledge spillover networks, and their geographical spatial patterns in different intra-/inter-regional contexts of China (Ma et al., 2022; Yang et al., 2024; Zhao et al., 2024), research regarding the GBA to date, largely reduces it to a “pre-given” political or geographical construct.

Theoretical framework

We draw on Henri Lefebvre’s (1991) work on the production of space and Ronald S. Burt’s (2000) social network theory to understand how social space is produced through a mutually constitutive triad (i.e., spatial practice, space of representation, representational space) and substantiated by pathways, networks, and clusters of relationships. Lefebvre (1991) advances a critical understanding of space as a social-political construct rather than a mere physical entity; and deeming the conceived as the dominant does little justice to the human agency within the “lived” or “representational” space. Burt (2000) lends a complementary lens to conceptualizing space as structure that creates competitive advantages for well-connected agents to accumulate social capitals from structural holes presented across the spatial triad, serving their projects and actions in the lived space. Their identities and imaginings are not situated within a pre-determined objective space structured by policies and geographical planning, but are in a constant “state of becoming” (Singh et al., 2007, p. 199). Figure 1 demonstrates the integrated framework.

Fig. 1 Theoretical framework: Lefebvre's perceived-conceived-lived triad of space and Burt's social network structure (source: the authors)



Spatial practice embodies associations within the perceive space between the “daily routine, routes, and networks” (Lefebvre, 1991, p.38) that link up the body that individuals mobilize energies with their places of private life and work (e.g., universities, intermediary institutions, companies in science and industrial parks). Representations of space (conceived space) is imaginary, real, and dominant in most societies, where the state, comprising of “scientists, planners, urbanists, technocratic subdividers and social engineers” (p. 38), establishes ideologies, policies, and directions of capital to shape the perceived and lived spaces of academia. The normative role of academia in cross-sectoral interactions as well as desirable and legitimate types of participation have been outlined in policies and regulations (e.g., MOE, 2017; MOST, 2015).

Representational spaces are the lived spaces of “inhabitants” and “users” and the usually dominated spaces that these agents seek to appropriate and change with their knowledge and actions (p.38). Key nodes exist in every network (i.e., government, industry, academia) that function with distinct institutional logics while interacting with each other. In representational spaces, they traverse institutional and city-region boundaries to mobilize collaborative opportunities in teaching, research, and valorization. These experiences develop capabilities and advantages in synthesizing different priorities and practices, particularly the “sticky” tacit information inside and across groups, while leveraging competitive advantages in their extant trusting and communicative “closure network” (Burt, 2021).

Spatial and social network theories throw significant light on human agency in interpreting and acting upon “the networks of social action” amid geopolitical and macroeconomic transformations (Singh et al., 2007, p. 198). However, the intricate understanding of network, space, and spatiality is sparsely represented in educational research (Danaher et al., 2003; Robertson et al., 2002) and is yet to be applied to studies of the GBA. The substantial opportunities that a GBA space implies and the limited knowledge about faculties as network brokers in producing such a space, together, inspire this study.

Methodology

To unpack how academics perceive, conceive, and Live the GBA higher education space, this research draws primarily on semi-structured interviews with 39 academics from Shenzhen, Guangzhou, Hong Kong, and Macau, the four city-regions positioned to be the international platform for “cross-border flows of innovation factors and regional integration” (State Council, 2019). Interviewees’ accounts are complemented by secondary data on funding schemes, institutional infrastructures, and the timing and sources of related investments.

Interviews were conducted in person and online between September and November 2023, lasting 1 h on average. Interviewees were recruited through purposive sampling and snowballing, spanning science, technology, engineering, mathematics (STEM) and social sciences and humanities (SSH). Purposive sampling involved analyzing faculty profiles, institutional websites, and media reports to identify individuals with cross-sectoral experiences (e.g., roles in government, industry, or civil organizations). Confirmed via questionnaire responses attached to invitation emails, thirty interviewees were identified as active “key nodes” and the subject of the analytical focus. Their profiles are given in Table 1.

Thematic analysis was conducted with each key node interviewee and then across different categories (e.g., SARs vs. Guangdong, brokerage with industry vs. Government or civil association) to generate inductive codes which were then combined with deductive codes informed by the theoretical framework. These codes were further integrated with complementary analyses of policies and statistics to identify key categories of venues, institutions, policies, and financial resources highlighted by the interviewees as crucial to their perception, conception, and construction of the GBA space. Patterns of relationships across themes emerged in this iterative process, which were then compared with data from the interviewees with “n/a” cross-sectoral engagements to derive supplementary insights into the key node attributes in perceiving and constructing the GBA space.

Findings

Perceiving the GBA space: policy-driven design, superimposing spatial practices, and emergent flows

For most interviewees, the GBA higher education space was initially and often instinctively conceptualized as constellations of existing and emerging venues in higher education and beyond, proceeding with ongoing investments, institutionalization, networks, and flows (e.g., MC-P1, MC-P2, HK-P4, HK-P5, GD-P1, GD-P2). These venues, varying in scale and function, were underpinned by the GBA’s Science, Technology, and Innovation (STI) and regional synergy strategy. They range from strategic cooperation zones (e.g., Hengqin, Hetao) and science parks with experimental institutional forms, preferential policies, and expanding R&D infrastructures to institutions that channel strategic focus and tangible resources into academic cross-sectoral activities and entrepreneurship.

The development of these venues reflects increasingly co-opetitive (i.e., concurrently cooperative and competitive) relationships between SARs and Guangdong in enhancing higher education’s synergistic capacity both independently and through cross-sectoral collaboration (MC-P13, HK-P2, HK-P7, GD-P1, GD-P4), albeit with notable asymmetries in focus. For many interviewees, the GBA higher education space was primarily manifested

Table 1 Cross-sectional information of participants

Participants (MC = Macau, HK = Hong Kong, GD = Guangdong)	Category of cross-sectoral engagements (by the time of empirical work)	Disciplines
MC-P1	Industry	STEM
MC-P2	Industry	STEM
MC-P3	n/a	STEM
MC-P4	Industry	STEM
MC-P5	Government and civil association	SSH
MC-P6	Government	SSH
MC-P7	Industry	STEM
MC-P8	Industry, government, and civil association	STEM
MC-P9	Government and civil association	STEM
MC-P10	Industry and civil association	SSH
MC-P11	Government and civil association	SSH
MC-P12	n/a	SSH
MC-P13	Industry and government	STEM
MC-P14	Industry	SSH
HK-P1	Industry and government	STEM
HK-P2	Hospital, industry, and government	STEM
HK-P3	Government	SSH
HK-P4	Government and civil association	SSH
HK-P5	Industry and government	STEM
HK-P6	Industry and government	STEM
HK-P7	Industry and government	STEM
HK-P8	Industry and government	SSH
HK-P9	n/a	SSH
HK-P10	n/a	STEM
HK-P11	n/a	SSH
HK-P12	n/a	SSH
HK-P13	n/a	SSH
GD-P1	Government	SSH
GD-P2	Industry	STEM
GD-P3	Industry	STEM
GD-P4	Industry	STEM
GD-P5	Government	SSH
GD-P6	Industry	STEM
GD-P7	Industry and government	SSH
GD-P8	n/a	STEM
GD-P9	n/a	STEM
GD-P10	Industry	STEM
GD-P11	Industry and government	SSH
GD-P12	Industry	SSH

through SAR universities' cross-border institutional presences in Guangdong, driven by the need to expand entrepreneurial opportunities and physical capacities, and supported by Guangdong's proactive supply-driven venue construction (e.g., MC-P1, MC-P2, MC-P14,

HK-P10, GD-P2). Meanwhile, the growing capacity of Guangdong universities intensified competition with their SAR counterparts for resources, talents, partnerships, and municipal support. In shaping the GBA higher education space, however, Guangdong universities focused more on the conceived space of the GBA through alliance-building and the strategic framing of collaboration (MC-P12, GD-P5, GD-P11, GD-P12).

Beyond a singular policy blueprint, the GBA serves an umbrella that facilitates both earlier practices with ad hoc institutionalization and sustained investments and the construction of new venues supportive of academic cross-sectoral activities (MC-P2, MC-P11, HK-P2, HK-P7, HK-P8, HK-P13, GD-P3, GD-P4, GD-P6). This concurrent superimposing and emergent characters are manifested in the periodic expansion of cross-border venues, university–municipality partnerships, and emerging institutional forms catalyzed by the GBA. As Table 2 indicates, various SAR university-affiliated institutions predated the formal launch of the GBA initiative. Initially established and funded by SAR universities and their foundations as intermediaries to the mainland, many increasingly gain financial support from local governments as “neo-type R&D institutions”¹ operating with flexible governance and orientation toward demand-driven R&D and commercialization.

Interviewees from both SARs and Guangdong highlighted the strategic role of these venues in addressing SARs’ constraints in industry scale, market demand, and physical capacity while meeting Guangdong’s quest for enhancing academic contributions to innovation-driven development. Notable examples include the Zhuhai MU Research Institute and HKU Shenzhen Hospital, which key node interviewees described as “official mediators” providing institutional endorsement for ongoing collaboration and entrepreneurial engagement in the mainland (e.g., MC-P1, MC-P4, MC-P5, MC-P13, HK-P2, HK-P7, GD-P4, GD-P6). As well as being strategic locations that facilitate favorable access to the mainland market and reduce operational and human resource costs, these venues are valued for their combination of the “operational autonomy” of SAR institutional identities and mainland legal status that streamlines administrative and contracting practices with mainland organizations and allows direct funding management to reduce financial loss from currency exchanges² (MC-P1, MC-P5, MC-P13, HK-P5, HK-P6, HK-P8).

The GBA also endorses the construction of “GBA-branded” institutions (e.g., the GBA university in Dongguan, GBA Institute for Smart Healthcare Innovation) that serve as platforms for alliance-building, resource channeling across sectors, and advancing collective GBA visibility (HK-P8, GD-P1, GD-P3, GD-P5, GD-P10).

Beyond the instinctive perception of the GBA as a constellation of venues and flows, interviewees also viewed it as a policy-driven sphere strategizing towards “mainlandization” and “STEM-ization” trajectory of STI development, supported by institutional endorsement, resource allocation, and coordinated central-local implementation (e.g., MC-P5, MC-P6, MC-P13, HK-P2, HK-P7, GD-P2, GD-P3). Its innovation-centric strategy (State Council, 2019) was operationalized through multi-level funding schemes supporting

¹ Since 2017 (prior to the publication of national guidelines from Ministry of Science and Technology), Guangdong Province has pioneered the issuance of documents that clarify the legitimacy of “neo-type R&D institutions” in government project participation, professional title evaluation, and eligibility for 5 million RMB construction funding from the Department of Science and Technology, along with subsidies for start-ups, equipment and R&D, with an annual funding budget of 150 million RMB in total matched by additional municipal funding (NDRC, National 2024).

² The Institute can receive funding from some mainland organizations, but not yet from the central government. For the official statement regarding Zhuhai Institute’s legal status, see [in Chinese]: https://hengqin.gd.gov.cn/zwgk/zdlyxxgk/spgg/content/post_4323374.html

Table 2 SAR university-affiliated institutions in Guangdong

Operation start time	Name of institution	Investment entities & amount
2005	Hong Kong City University (CityU) Shenzhen Research Institute	CityU: 72.8 million RMB
2006	Hong Kong University of Science and Technology (HKUST) Shenzhen Research Institute	HKUST: 72.8 million RMB
2007	HKUST Fok Ying Tung Research Institute	Fok Ying Tung Foundation: 1.6 million RMB
2010	Hong Kong Polytechnic University (PolyU) Shenzhen Research Institute	PolyU: 87.435 million RMB
2011	HKU Shenzhen Institute of Research and Innovation	HKU: 50.14 million RMB
2011	CUHK Shenzhen Institute	CUHK: 22.4 million RMB
2012	Hong Kong University (HKU) Shenzhen Hospital	Shenzhen Municipal Government: 6 billion RMB
2019	Zhuhai Macau University (MU)—Science and Technological Research Institute	n/a
2021	Zhuhai Macau University of Science and Technology (MUST)—Science and Technological Research Institute	n/a
2022	HKUST (Guangzhou)	Guangdong Provincial & Guangzhou Municipal Governments: 225.56 billion RMB
2024	HK CityU (Dongguan)	Dongguan Municipal Government: 2.1 billion RMB; enterprise Donation: 22 million RMB
2026	Chinese University of Hong Kong (CUHK) Shenzhen Hospital	Shenzhen Municipal Government: 5.1 billion RMB

Source: Prepared by the authors based on information from official government and university websites (as of December 2024)

academia–industry collaboration. As shown in Table 3, some of these schemes target cross-border partnerships—primarily led by mainland governments—while others focus on STI capacity-building and require enterprise leadership and proportional co-funding.

Despite the central ambition to forge a unifying vision, academics, the actual “inhabitants” of this space (Lefebvre, 1991), p.39), remain agentic in choosing their forms of engagement and network-building according to intellectual and professional interests. For many who chose not to, or had yet to develop the capacity to engage with the regional vision, the GBA remained a policy-driven conceived space, distanced from lived experiences. The interstices between the perceived, conceived, and lived spaces were illuminated among interviewees who concentrated on teaching and research, who believed that the government “expects too much” from academics in fulfilling the “third mission” (HK-P9, GD-P8, GD-P9), and among SAR academics in the social sciences and humanities, who revealed few incentives for valorization through cross-sectoral engagement but concerns over compromise of intellectual interest when negotiating external priorities—tensions further compounded by potential authoritarian control over academic freedom in mainland networks.

Foreign academics also viewed themselves as “outsiders” to the GBA initiative, citing unresolved barriers related to language, national identity, and social networks and noting that the perceived benefits accrued mainly to ethnically Chinese academics (HK-P11, HK-P12, HK-P13). Many of these SAR academics, both local and foreign, perceived themselves as structurally autonomous in network-building, neither bound to choose between local and (sub)national engagement nor compelled to collaborate with industry or government as valorized in the GBA’s conceived space. Even among those typically seen as “direct beneficiaries” of such valorization (e.g., academics in mainland and STEM), some also revealed self-sufficiency with their existing local or international networks for advancing research and entrepreneurial goals without aligning with the GBA initiative.

Overall, the GBA was perceived by most interviewees as a policy designated space that superimposes on existing constellations of spatial practices, networks, and flows—periodically incentivizing their emergence through ad hoc strategies and support, and imposing “reciprocity” on valorized ways of production (Lefebvre, 1991, p. 56). However, the presupposing use of spatial practices is not fully cohesive with the practical demands of everyday life. For instance, while government statistics presented the Hengqin Zone as populated by 5763 Macau-invested enterprises and 7473 Macau residents (Economic Development Bureau of Hengqin, 2023), SAR key nodes perceived it as lacking community “vibrancy” and *de facto* industries. MC-P13, a U.S. returnee who founded a company in Hengqin for instance, noted that border crossings remain bureaucratically challenging for foreign passport holders and physically uncomfortable.

For interviewees who did not to (actively) engage in cross-sectoral practices, policy and industrial production spheres remained distant from their everyday routine of teaching and knowledge production (MC-P3, MC-P7, MC-P12, HK-P9, HK-P10, HK-P11, HK-P12, HK-P13, GD-P8, GD-P9). Yet the talents they cultivated and retained, and the knowledge they generated, which influenced productivity directly or indirectly, constitute representational spaces of the GBA in their own right. Their representational spaces are not dominated by the ideology or political apparatuses of the GBA initiative due to the strengths of individual agency and the intellectual autonomy or academic freedom afforded by their universities.

Academics in key node positions bridging academia, industry, and government networks tend to strategically appropriate and even help construct the GBA space. They do so by “decoding” the conceived space of its ideologies, arbitrage, and constraints, translating

Table 3 Examples of funding schemes supporting multi-sectoral collaboration

Funding body	Title of funding scheme		Application requirements and funding scale
Central government (Guangdong Science & Technology Department, 2023a , 2023b)	Key Special Programs of “Strategic Scientific and Technological Innovation Cooperation”: HK, Macau, Taiwan Projects	Enterprises lead or jointly apply with research institutions. At least one enterprise must participate and provide Matching funds. Expected total: 30 million RMB	
HK Innovation and Technology Commission ITC & MOST (n.d.)	Mainland-Hong Kong Joint Funding Scheme	University-enterprise joint application, with industry sponsorship covering at least 10% (platform projects) or 50% (collaboration projects). Remaining costs co-funded by ITC and MOST	
Guangdong Province & Prefecture-Level Cities (Guangdong Science & Technology Department, 2023a , 2023b)	Guangdong-Hong Kong-Macau Science and Technology Cooperation	Guangdong enterprises lead application jointly with HK/Macau universities and research institutes. Single project funding ranges from 200,000 to 1 million RMB	
Guangzhou Science and Technology Bureau (2023)	University-Enterprise Joint Funding Project	Application is open to universities, research and healthcare institutions, and enterprises in Guangzhou. Maximum total funding 60 million RMB	
Shenzhen Science and Technology Innovation Committee (Shenzhen 2023)	Shenzhen Major Science and Technology Projects	Application is open to Shenzhen-based enterprises, universities, and research institutes. Maximum 10 million RMB per project	
Hong Kong Innovation and Technology Commission (n.d.a. , n.d.b.)	Partnership Research Programme	Matching funds for collaborative R&D projects, with industry sponsorship covering at least 50% of the total cost	
Macau Science and Technology Development Fund (2024)	Enterprise-University-Research Institute Matching Grant University-Enterprise Innovation Resource Matching	Application is open to Macau enterprises. Maximum total funding 25 million MOP Joint applications from Macau enterprises and universities. 3–5 MOP million per project	

Source: the authors compiled from information on official websites of the listed funding bodies

them into concrete interpretations and practices in the “language,” “mindset,” and “interest” of colleagues and partners (MC-P1, MC-P5, MC-P8, MC-P13, HK-P2, HK-P3, HK-P5, HK-P6, HK-P7, GD-P3, GD-P6, GD-P7, GD-P11). The diverse perceptions, projects, and actions—whether to engage with the GBA or not—have been crucial in counterbalancing the homogenizing effects of political power and fertilizing diverse forces of production in and for the GBA higher education space.

Key node academics: plural networked strategies and cross-fertilizing pathways in appropriating and constructing the GBA space

Unlike the “split personalities” often experienced by academics in cross-sectoral practices, key node academics effectively synthesized the priorities and attributes of academia, government, and industry, including (semi-)public institutions such as hospitals and research institutes, into their roles and day-to-day practices. They were particularly alert to the opportunities associated with the GBA initiative and proactive in fostering academic projects through a swirling appropriation of venues, institutions, financial resources, and connections, aligning with valorized modes of production (e.g., “SARs R&D + mainland commercialization”) and strategic maneuvering of priorities across-sectoral networks.

To “synthesize resources for greater initiatives,” many spearhead the production of new infrastructures, platforms, regulatory frameworks, and funding schemes, conceiving their forms and functions with synthetic expertise and resources from colleagues and industrial partners (MC-P8, MC-P13, MC-P14, HK-P1, HK-P2, HK-P5, HK-P6, HK-P7, GD-P3, GD-P6). Projects aimed at producing applicable achievements have brought academia out of the abstract space of “theoretical knowledge” closer to the lived space of everyday life and production activities. More importantly, their direct “occupation” of the GBA space (Lefebvre, 1991, p. 179, 190) served to bring the predominantly abstract space of GBA blueprint and policies closer to the day-to-day realities of academia and production.

However, imperatives for cross-sectoral engagements and the understandings of the *raison d’être* behind these venues, funding, and appropriation strategies also vary between SARs and Guangdong academics and by academic roles, disciplines, citizenship identities, and individual experiences. SAR interviewees, facing immediate local capacity constraints, tend to articulate the motives and implications of appropriating cross-border venues, whereas Guangdong academics often perceive them as expanded, but not necessarily primary choices. Key node academics who establish their own ventures or hold leadership roles in developing spatial practices for their institutions also incline to articulate “why” and “how” the GBA space can be effectively appropriated. The following sections compare how key node academics appropriate and construct the production sphere of GBA through plural strategies and pathways.

Key node academics in the GBA production sphere

For key nodes engaging in industrial networks and entrepreneurship, the economic and business realms were perceived as the most vibrant and consensus-driven across the GBA that offer a promising departure for implementing the GBA initiative (MC-P1, MC-P2, MC-P4, MC-P7, and other 17 participants). By integrating academic knowledge into production, they enhance productivity through industry partnerships, commercialization, spinout creation, and entrepreneurship education. Their deep-seated commitment to innovation serves as a key engine of steering existing modes of production toward alignment

with the STI strategy. These engagement addresses the practical needs of GBA “inhabitants” (including markets and end-users) while contributing to the region’s development as a polycentric “forcing house” of capital accumulation (Lefebvre, 1991, p. 53), capable of attracting incoming inflows of talent, business, and capital. Such engagement unfolds through different professional roles, disciplinary specializations, and pathways across SARs and Guangdong.

Industrial partnerships, venture creation, and entrepreneurship education

By taking on roles as “industrial node” or “entrepreneur,” key nodes harness funding opportunities targeting industrial innovation and commercialization (see examples in Table 3). For full ownership and operational autonomy, some established companies in incubators or science and industry parks to leverage the benefits of subsidies, tax incentives, reduced rents, and/or preferential policies for personnel, resources, and capital exchange in SAR-mainland cooperative zones; others amplify these benefits by operating companies on both sides of the border (MC-P13, GD-P6, GD-P8).

With valorization of academic knowledge as the central priority, they manage early-stage operational stability and financial viability of enterprise growth while persistently capitalizing on academic projects. Despite undervalued prices, they consistently support technology transfers to industry in order to achieve growth scales not possible within academia or hospitals alone. They decisively withdraw when shareholders’ focus on extracting profits from existing intellectual properties (IPs) compromises their “deep-seated passion” for innovation, which they see as undermining a deeper commitment to innovation, scientific advancement, and broader societal impact (MC-P1, HK-P2, HK-P7, GD-P6).

Beyond supporting enterprise growth, industrial partnerships and spin-outs also provide entrepreneurship education and employment opportunities for graduates, cultivating future productive forces—labour, knowledge, and enterprise—for the GBA (MC-P5, MC-P8, MC-P9, HK-P5, HK-P6, GD-P3). Academics engaged in entrepreneurship education see themselves as playing a pivotal role in guiding students to incubate projects that contribute to the GBA’s social, economic, and cultural development (MC-P4, MC-P14, HK-P4, GD-P11, GD-P12).

Disciplinary specializations: STEM-ization and mainlandization

Despite disciplinary diversity in how academic engagement enhances productivity, the production sphere is largely characterized by STEM-ization—with most participants from STEM fields—and mainlandization, reflecting a dominant flow toward integration into mainland markets. Academic engagement remains strongly oriented toward the mainland, driven by arbitrage in access to a vast market, expansive physical space, comprehensive R&D infrastructure, preferential policies, and a complete supply chain.

Given the infrastructure and funding-intensive nature of STEM and medical sciences, key node academics in these domains tend to act as early movers, pre-occupying spatial practices (e.g., new R&D facilities) and securing multiple industrial funding. Their engagements contribute materially to the construction of the GBA through tangible outputs such as public infrastructure in civil engineering (HK-P5) and smart city and power supply systems in electrical engineering (MC-P2, MC-P4, MC-P8, HK-P7), with some extending their influence into the conceived space through system design, standard-setting, and cross-border policy coordination.

In contrast, the few academics from the social sciences and humanities engaged in the production sphere are primarily urban planners and architects (MC-P10, HK-P8, GD-P7), who focus on shaping spatial configurations through consultancy-based knowledge exchange with industrial partners. Interpersonal connections and the strategic value placed on academia-industry collaboration often carry more weight than spatial practices.

For those in medical and pharmaceutical industries, working across borders involves strategic navigation of plural markets and legitimate sites to maximize value from academic projects. As HK-P2, MC-P1, MC-P13, and GD-P6 indicated, despite aligning closely with international standards by importing products registered by foreign agencies, HK and Macau, as SARs, lack independent regulatory bodies to approve clinical trials and register new drugs. Except for a few pilot organizations, SAR medical products' entry to the mainland requires approvals from China's National Medical Products Administration. By working with and for medical companies and hospitals across borders, key nodes enhance the provision of GBA's healthcare with higher medical quality and efficiency not only through the offerings/products developed from academic knowledge, but also through cultivation of medical professionals and coordination of medical institutions and resources involving governments across SARs and Guangdong, which will further discuss in the next sections.

SAR key nodes: pre-emptive occupation of collaborative pathways to Guangdong

SAR academics increasingly seek to build first-mover advantages in yet-to-be-populated space in Guangdong and to secure research facilities from municipal governments before the issue of official plans—for instance, for the Hetao Shenzhen-Hong Kong Science and Technology Innovation Cooperation Zone. The practical demands for convenient commuting, institutionalizing bottom-up interactions, and legitimate cross-border mobility of personnel and financial resources that the brokering institutions addressed bring the conceived space closer to the perceived and lived spaces of academia and industry.

As SAR academics appropriate drives and energies for in-person interactions with mainland partners, relocating their work and leisure to these venues, more industries and “inhabitants” are drawn to these lived spaces (MC-P1, MC-P8, MC-P9, MC-P13, HK-P1, HK-P2, HK-P7). This allows the GBA conceived space to gain concrete abstractions and reinforce control over the form and function of particular areas, such as Hengqin, in becoming a cooperative “industry-university-research base” through the “Macau R&D + Hengqin Commercialization” pathway (Guangdong Provincial Government, 2024; National Development & Reform Commission, 2023a, 2023b).

Those in managerial roles meticulously promote the benefits of multiplying project funding from municipal governments and the operational strategies for utilizing mainland funding, samples, and data according to changing policies and practices. The academic expertise, institutional advantages, and reputation of SAR universities are leveraged to facilitate design of research facilities, educational programs, commercialization pathways, and multi-departmental collaborations in SAR-affiliated institutions. The social capital of both, alongside their successful pilot practices are subsequently pitched to municipal governments for endorsements and funding to develop new facilities and programs. In initiatives involving coordination of regulatory frameworks, SAR hospitals and governments become the closure network to facilitate the institutional status and pioneering practices of SAR-affiliated institutions.

Guangdong key nodes: retraction and discursive strategies

For Guangdong interviewees, the mega-construction of spatial practices appears as given conditions for social practices. They reveal higher expectations for commuting efficiency and less commitment to border-crossing than their SAR counterparts, which is compounded by increasing cross-border restrictions on mainland academics with administrative roles on official occasions (GD-P9, GD-P10). Those who co-initiated companies or projects in HK remain based in Shenzhen and delegated operations to HK partners (GD-P2, GD-P6, GD-P8).

While SAR academics frequently mobilize university endorsements for multi-sectoral collaborations, such favorable institution is less an established fact for Guangdong academics, who need to strategically seek them out by choosing Special Economic Zones and universities offering greater autonomy for cross-sectoral practices. This is a key reason why many mainland key node interviewees opt for SAR universities or those with similar systems (GD-P2, GD-P4, GD-P8, HK-P1, HK-P5, HK-P6, MC-P10, MC-P12). Navigating university regulations for entrepreneurial activities requires discursive practices where the adage “non-opposition as consent” prevails as strategic maneuvering over ambiguities and tensions between central directives and university practices:

The State Council has issued documents encouraging academic entrepreneurship, but many universities don't. They see it as a distracting side mission. Our university's policies are uniquely more flexible than those of many others in Shenzhen and across Guangdong. (GD-P8)

While key nodes generally have established cross-sectoral networks in their city-regions, Guangdong academics tend to consider local networks as self-sufficient and are less compelled by the “strategic importance” of collaborating with SARs, especially given the lack of relevant industries and demands, alongside perceived distance and constraints in reach (GD-P2, GD-P3, GD-P4, GD-P7, GD-P9, GD-P10). Increasing collaborations between mainland industries and SAR academia lead to potential “dilution” of their social capital, though this can be a “reshuffling” that optimizes talents and resources, forging a “latecomer advantage” in cultivating innovation capability: “Those who withstand the reshuffle will stay, continuously replenishing new blood and talent to Shenzhen, where its main advantage over HK lies” (GD-P4).

Key nodes remain agile to the changing connectivity and competence of city-regions. Reshuffling potentially also shifts perceptions of where rewarding opportunities burgeon and thus efforts to redirect to. In contrast to the broader trend of mainlandization among SAR key nodes, GD-P8, who co-established a company in Shenzhen, drawn by earlier subsidies and incentives for HK entrepreneurs, now sees more lucrative opportunities in establishing a branch office in HK. This shift is driven by the HK government's increasingly proactive policies and funding for industrial development and talent retention, contrasted with a diminishing confidence in Shenzhen's fiscal capacity to maintain these benefits post-pandemic.

Engagement with intergovernmental networks and the conceived space

While industrial engagement follows a straightforward economic logic, engagement with government networks, the predominating conceived space, involves complex strategies and tensions in shaping the modes of production and production relations. Pathways to

influence the conceived space and bridge governmental interactions vary according to institutional roles, strategic objectives, and network positions.

For those primarily taking industrial roles and specializing in STEM, brokerage is embedded in the institutional interests and practical demands for development, which is also eminent in SAR academics' engagement with government networks (MC-P1, MC-P2, HK-P3, HK-P4, HK-P5, HK-P6, GD-P7). SAR key nodes appear notably entrepreneurial in cultivating distinctive niches within cross-border networks by facilitating pre-emptive collaboration pathways. They remain proactive in brokering intergovernmental connections between the SARs and Guangdong and establishing a regional governance layer that addresses the long-term demands of academic cross-sectoral practices and cross-border mobility while pitching mainland industry to SARs governments (public procurement and supply, policymaking capacity) and industry (enhancing production capacity).

Those who position themselves as "public intellectuals" or "scholar-officials," representing the academic voice in government decisions, tend to influence policymaking through intellectual and ideological contributions under the ethos of "studying to become officials" (MC-P3, HK-P1, HK-P2, GD-P1, GD-P2, GD-P6). They primarily enhance the GBA's productivity through the strategic positioning of higher education in driving regional development and the scientific planning of population growth, land use, infrastructures, and institutions that optimize their utilization. Nevertheless, these pathways often appear hybridized, particularly for those taking managerial roles.

SAR academics on the "less-trodden" pathways: reshaping production relations

SAR key node academics, particularly those with mainland networks (MC-P6, MC-P8, MC-P9, HK-P7) or hybrid identities as mainland-trained scientists working in SARs (HK-P2, HK-P5, HK-P6, HK-P8, MC-P13), possess distinctive agency in reshaping the production relations by building cross-fertilizing networks across the academia, industry, government and society of SARs and mainland and a long-term regional governance structure.

From intergovernmental brokerage to trilateral regional governance

By brokering local government networks in case-by-case negotiations and collaborations in long-term agenda-setting, policymaking, and producing spatial practices that cross-fertilize academia and industries of SAR and Guangdong, SAR key nodes coordinate diverse political targets and strategies towards mutual interests in high-profile economic and innovation activities. However, some local tensions necessitate orchestration by the "higher power" of the central state, particularly amid the GBA's legal, institutional, and social-cultural heterogeneity. Hence, some seek direct endorsements from the central government for national projects as well as for construction of key infrastructures and frameworks with successful local initiatives (MC-P6, MC-P13, HK-P2, HK-P5, HK-P6). These central endorsements not only serve as political achievements and promotion opportunities for local government officials, but also enhance the "exposure" of key-node academics within local networks, which opens up rewarding opportunities (e.g., additional funding, advisory positions in local governments).

This "oscillation" within the central-local nexus also conduces to establishing an intermediary layer between the central state and provincial/municipal governments. Beyond case-by-case endeavors, key nodes such as HK-P5 and MC-P6 are striving to transform the

Hong Kong-Zhuhai-Macao Bridge Authority into a regional GBA governance structure, a permanent establishment that would enable efficient, long-term coordination of regulatory, institutional, and ideological differences. With administrative functions extending beyond transportation and talents from all three regions, this establishment would not only optimize the Bridge's strategic potential, but also create opportunities for the institutional breakthrough of one country, two systems.

Pitching mainland industry to SAR governments and industries

While “SAR academia + mainland industry” is succinctly facilitated by SARs and mainland governments, “mainland industry + SAR governments” remains a pathway in question, pre-emptive of the national strategy while challenging HK's path dependency on international suppliers. Advocacy for mainland industry's equal participation in SARs' bidding challenges such hierarchical preferences and “exclusion” stipulated by SAR governments, demanding substantial commitments to accumulate reputational trust and re-shape “mindsets” (HK-P5). Commitments are particularly eminent among scientist/engineer SAR academics who have substantial experience in SAR industries and mainland governments or quasi-government positions (e.g., the Bridge Authority, leadership of state-owned enterprises). They strive to help SAR governments and industries “seek the best” with extensive referrals on potential suppliers, partners, and financing across the GBA (HK-P5, HK-P6, MC-P6, MC-P8).

This less trodden yet highly rewarding pathway allows them to establish a strong foothold in the SARs with competitive advantages unparalleled by mainland or SARs counterparts and, more importantly, produce a new equilibrium of development for academia, industry, and governance across the GBA, and even internationally. It is also expected to enhance HK's capacity to develop endogenous solutions and technologies in collaboration with mainland industry. As a HK key node, who has been diligently promoting mainland industrial capability to HK and beyond explained:

HK is used to delegating construction projects to foreign consulting firms and contractors, meaning that technologies remain in others' hands....We [mainland] can deliver the same standards much cheaper and faster....That's why I keep persuading HK SAR officials into changing mindsets and allowing mainland manufacturers to participate equally in HK's project bidding.... This benefits government projects and the application of advanced mainland industrial achievements in HK. More importantly, it provides academic with problems to work on. (HK-P5)

By assuming advisory roles in government committees, think tanks, and civil associations and serving as members of the Chinese People's Political Consultative Conference and SAR legislative councils, key nodes leverage various participatory channels to enhance opportunities and social capital in shaping the organization of space—strategizing where and how capital and resources are allocated. SAR key nodes stay attuned to the political importance of integrating SARs into national development and the notable influence of mainland strategic planning traditions on SARs' strengthening STI deployment, which is also reflected in the increasing funding schemes established under the GBA framework (Table 3). Accordingly, they are committed to developing ongoing connections with mainland officials through “paying respects” (拜码头), “dining,” “drinking,” and sometimes partnering with mainland academia to facilitate appropriate cultural practices and trust (MC-P9, MC-P13, HK-P3, HK-P4, HK-P7). As a native HK faculty noted:

The Party-state explicitly expects HK, Macau, and Taiwan to join space programs as the United Front...Although mainland technologies might not be as advanced as HK's at the time, if they didn't trust our commitment to the country, they wouldn't have included us.... Decisions often hinge on central leadership's approval. (HK-P7)

Scholar-officials in Guangdong: embedded in policy and ideological space

Multiplying academia-government interactions reveals one-sided directionality. Despite SAR academics' endeavors to be "heard" by both SARs and mainland governments, Guangdong academics remain exclusively proactive to the central directive for organized scientific research and its changing policy hotspots, with the GBA now just one among many (GD-P1, GD-P5, GD-P9). Discrepancies between scholarly input and final policy documents are often seen by SAR academics as reflecting differences in priorities and "timetable" to get used to (MC-P6, MC-P8, MC-P13, HK-P4, HK-P8), whereas Guangdong academics remain sensitive to the mainland government's ideological emphasis on scientific policymaking and academia's role in shaping "policy ideas" ("思想") and "discourses" ("提法") as well as in combining "scholarly theoreticality" ("学理性") with cutting-edge examples from research in cadre training (GD-P5, GD-P9). This policy-driven tradition has formed a self-sufficient intellectual community, deemed epistemically distanced from SAR academia, particularly in social sciences and humanities (HK-P3, HK-P4, HK-P9, MC-P12). However, it is appropriated by Guangdong academics as a "professional" source of learning how consensus codes are established through policies, particularly where "compromises" are made for the sake of "feasibility" in order to improve their own capabilities to re-shape the conceived space (GD-P1, GD-P5).

Discussion and conclusion

Producing a GBA social space: mutually constitutive triad and cross-fertilizing networks and pathways

This study advances spatial theories with empirical insights into how the perceived-conceived-lived triad mutually constitutes the production of GBA social space where key node academics, as connected agents, facilitate a coherent whole of tripartite space by establishing a common language, consensus, and appropriate code of practices that synthetically address the properties of academic, industry and government networks (Lefebvre, 1991, p. 40). In line with Lefebvre, spatial practices are often directly "lived," extending the vicinity of body into building surrounding networks and pathways in SARs and bordering city-regions in Guangdong. By addressing some practical demands through the communal use of spatial practices, the conceived space brings itself closer to the lived space of academia and industry. Key nodes' lived space draws industries and incoming inhabitants to populate where they are devising energies around. This allows the GBA conceived space to gain concrete abstractions and reinforce control over valorized forms and functions of spatial organization.

The GBA conceived space serves to structure the organization of space and regularize flows and capitals, leading to homogenizing tendencies; for instance, towards centralized strategic deployment driven by STI, which the state dictates as central. The commercialization of academic knowledge, particularly through patenting, IP production, and spin-out creation, continues to receive the most intense scrutiny by scholars and policy-makers

(Abreu & Grinevich, 2013; Guangdong Science and Technology Department, 2024; Jain et al., 2009; Jensen & Thursby, 2003), notwithstanding the diverse pathways of knowledge transfer such as contract research, consultancy, and joint training (Bekkers & Freitas 2008; Zhao et al., 2020).

However, the conceived space cannot always pass as established knowledge to conceal “reality,” nor can its goal of homogenization overshadow the diversity of lived experiences. Key nodes are often active users and producers of venues, ideologies, policies, regulations, and funding to reinforce control over their primary interests. With lived experiences holding significant sway in the day-to-day realities of academia and industry, they remain reflexively conscious about the arbitrage of the perceived-conceived space in supporting sustainable and cost-effective allocation of finite time and energies to multiple roles. Rather than sidestepping the “oscillations” of the conceived space, key nodes, the “small elite,” embrace them as ongoing practices to foster intergovernmental collaborations and establish a regional GBA governance layer as permanent centers for decision-making. At the organizational level, strategic maneuvering and discursive practices are also employed to navigate tensions between state endorsements and *de facto* university regulations for faculties’ cross-sectoral activities, which are not entirely within the state’s political space, as Lefebvre stipulated (p. 281).

This study also throws light on the contextualized relations of the perceived-conceived-lived triad of the GBA, which are never simple, stable, or generalizable across contexts. The triad contributes in different ways to the production of space according to their individual “qualities and attributes,” “mode of production in question,” and “historical period” (p. 46), and their relative weight varies across sectors and specific practical requirements. While perceived-lived cohesiveness carries notable weight in academia’s industrial engagements, the conceived space retains a strong hold in the government. Ideologies play an even larger role in academic engagements that aim beyond endorsements for practical demands to shape policies and governance. The conceived and lived spaces are not always in tensions or dominating-submissive relationships. Projects aimed at producing applicable achievements have brought academia out of the abstract space of “theoretical knowledge” closer to the lived space of everyday life and production activities. While cross-cutting the institutional boundaries (Etzkowitz & Zhou, 2019), key node academics “restore society in its entirety,” which was fractured “in highly complex fashion into models of sectors” and “systems” as an established product of objective analyses through the conceived space’s apparatuses, notably as innovation system and triple helix approaches (Lefebvre, 1991, p. 311).

Network entrepreneurs in higher education

This study contributes to an expanding conceptualization of structural holes beyond relational networks to include the spatial triad as a structure that allows network entrepreneurs to accumulate social capital through multiple pathways and directionalities. Attesting to Burt’s (2000) profile of network entrepreneurs, key node academics’ network structures make them early strategic movers in mobilizing productive opportunities and bringing them to fruition with synthetic resources and cross-fertilizing networks. Despite some similarities (e.g., being “risk-taking” or “venture-enjoying”), the business contexts that Burt’s theory built on do not fully explain the hybrid identities of academics, entrepreneurs or government advisors. Key node academics remain “enterprising” in pursuing social prestige rather than being profit-driven, where “academic heartland” holds centrality

(Marginson & Considine, 2000; Clark, 2001). Synthetic resources attained from cross-sectoral practices are fed back into teaching, research and public service.

Key node status is strategic, imagined and relational. Sustainable synthesis of multiple roles by leveraging positional advantages and resource repertoires (e.g., establishing entrepreneurship that aligns with academic evaluation criteria while attracting governmental attention) is a crucial quality as multi-sectoral brokerage demands substantial time, energy and sociability. Key node academics navigate and manage differences not just in “understanding across groups” (Burt & Burzynska, 2017, p. 6), but also in the institutional priorities fundamental in determining how information, resources, and practices can be mobilized, and which are further complicated by the institutional and cultural heterogeneity of the GBA. However, imagined status does not always fully align with relative positioning in network structures: what constitutes a “key node” from self-positioning still warrants further discussion: expectations of key node qualities vary across individuals (e.g., “reputation,” “social status,” “connection”).

Context-specific agency in brokerage is also under-explored in Burt’s work, where, under the influence of Confucian tradition, academics bear a distinct moral responsibility to “give back” to the public good rather than indulging in “competitive” brokerage. Echoing Lefebvre (1991), and Han and Ye (2017), the increasing public service role of science in administration and production has enhanced the status of scholar-officials, industrial key nodes, and “professional interest groups” as elite policy-makers. This warrants further research into how the Confucian tradition of intellectuals influencing public policy through “intellectual authority” and “self-mastery” manifest and carry weight in the GBA, where academic projects align with the Party-state’s “ideological interest and mandate,” along with the Euro-American traditions of academic freedom and institutional autonomy that customarily have a stronger hold in SARs (Han & Ye, 2017; Li, 2020; Marginson, 2022).

Policy implications

High-order spatial planning, combined with multi-scalar bottom-up initiatives and practices, have fostered a good many venues, infrastructures, resources, and institutions supportive of multi-sectoral network-building and innovation practices. It remains essential to both continuously recognize and support the pioneering endeavors of “ideologues,” “technocrats,” and “architects” in constructing the GBA higher education space and conceiving its future trajectories while tailoring strategies to address the disparate lived experiences within the GBA. The conceived space wields considerable primacy over producing spatial practices and normalizing the lived space on the mainland compared to HK and Macau. Guangdong sits between the rest of the mainland and the SARs with its relative openness, autonomy, and entrepreneurial culture as the demonstration zone of China’s Opening Up and Reform – attributes that make it promising for fostering regional synergy and innovation of the GBA.

While HK and Macau were once conceived as uniform in SAR policies and national agenda, the GBA strategy increasingly seeks to facilitate the synthetic strengths of diversified productive forces and differentiated positions in production, such as Guangdong’s advanced manufacturing and high-tech industry clusters, HK’s international finance, and Macau’s exhibitions and tourism (Development and Economic Bureau of the Guangdong-Hong Kong-Macau Greater Bay Area, 2023; State Council Liaison Office for Hong Kong and Macao Affairs, 2024). The future transformation of the GBA from the political imaginary of an “international first-class bay area” (State Council, 2019) into a social

reality largely depends on how these heterogeneous strategies, modes of production, and productive forces synergistically contribute to a distinct model of institutional, economic, and social innovation. The cross-fertilizing multi-sectoral networks within and beyond the GBA hold significant potentiality for enabling “domestic circulation as the mainstay, with domestic and international circulations mutually reinforcing” (NDRC, (2023)).

Although grounded in a non-Western context, this study offers broader insights for global settings seeking to strengthen the role of academia in STI-driven development and regional integration. Many of the challenges and opportunities associated with academic cross-sectoral engagement are not unique to the GBA. Amid global economic slowdowns and rising protectionism, there are growing expectations for higher education to take on expanded roles—fostering entrepreneurship, responding to government and market demands, and enhancing the synergistic capacity of local and national innovation systems. It remains critical to understand how academics, varying in intellectual and professional interests, continually reinvent their roles and capabilities to reconcile these external demands with the traditional academic core amid converging forces toward STEM-ization in achieving the STI-driven strategy as well as strategically navigating centrally policy-driven spatial configuration and local entrepreneurialism to advance their projects, while managing the tensions and co-opetition amid urbanization and state-rescaling in subnational regionalization.

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Declarations

Ethics approval and consent to participate This study was approved by the Ethics Committee of Southern University of Science and Technology. All participants provided informed consent prior to inclusion in the study.

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